Forecasting Fantasy Premier League Points: A Walk-Forward Evaluation of Captaincy and Lineup Decisions

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Abstract

Fantasy Premier League (FPL) managers choose, every week, eleven starters from a fifteen-player squad and a captain whose points are doubled. We forecast each player's points for the coming gameweek with gradient-boosted regression trees trained on four seasons of player-level match data (2022/23–2025/26; 79,885 player-gameweeks), using only information available before each deadline. The model is evaluated walk-forward over 114 gameweeks (2023/24–2025/26), retrained every week on all earlier weeks, and its decisions are compared with fixed baselines under an acceptance rule stated before the first run. The forecast is more accurate than recent form (mean absolute error 1.42 versus 1.50 points; correlation 0.51 versus 0.41). As a captain rule it outscores choosing the most expensive attacker who started the previous week by +1.54 points per week (95% bootstrap interval +0.45 to +2.61) and choosing the best recent form by +1.44 (+0.27 to +2.62), but trails the captain most often chosen by the managers who finished the season in the overall top 100 by −1.46 (−2.40 to −0.55). Selecting starting elevens from those managers' actual squads, the model beats a form rule by +2.39 points per week but trails the managers' own choices by −2.16. Ablations show which inputs carry the forecast, and the conclusions are stable across model settings. We also test whether a player's share of his team's possession-ending actions, an adaptation of basketball's usage rate, adds information. We discuss why the strongest baseline is both informative and biased, and why single-season evaluations cannot separate decision rules that differ by less than about one point per week.

1 Introduction

Fantasy Premier League is the official fantasy game of the English Premier League. Each manager holds fifteen players within a fixed budget and, before each of the season's 38 gameweek deadlines, names eleven starters and a captain whose points are doubled (Fantasy Premier League, 2026b). Players score for playing time, goals, assists, clean sheets, saves and bonus points and, from 2025/26, for defensive actions. Two decisions recur every week regardless of transfers: who starts, and who captains. Both reduce to ranking players by expected points in the coming week.

A small literature forecasts FPL points with statistical and machine-learning models. Early work treated team selection as sequential decision-making under uncertainty (Matthews et al., 2012). Later studies forecast points from recent match histories using time-series and recurrent models (Gupta, 2019), convolutional and recurrent networks over a player's last five matches (Lombu et al., 2024), and ridge regression, gradient boosting and convolutional networks (Frees et al., 2024). Others add information beyond match statistics, such as news and betting markets (Bonello et al., 2019), or couple forecasts with mixed-integer squad selection (Ramezani and Dinh, 2025). These studies mostly report forecast error or season-long team totals. Fewer ask whether forecasts make better *weekly decisions* than the rules and expert consensus managers already use, and fewer fix the criterion for success before looking at the results.

This paper asks that question. We are explicit about two things that often go unstated in sports forecasting: information leakage, meaning the use of data that would not have been known at decision time (Tashman, 2000), and the noise of weekly outcomes, which limits what any evaluation can show. Our contributions are:

1. a deadline-safe feature set that carries players' form across seasons and uses the previous season's rates as a prior, following the logic of empirical Bayes shrinkage (Brown, 2008; Efron and Morris, 1975);
2. a walk-forward evaluation that retrains every week and compares *decisions* against fixed baselines, with an acceptance rule set in advance (Nosek et al., 2018);
3. lineup evaluation on 11,400 real squads, with substitutions applied by the game's rules;
4. ablation and sensitivity analyses, so that results do not rest on unexamined choices; and
5. a test of whether ball usage, adapted from basketball's usage rate (Basketball-Reference, 2026; Oliver, 2004), adds information.

2 Data

2.1 Player-gameweek records

We use official FPL data for four complete seasons, 2022/23 to 2025/26 (Fantasy Premier League, 2026a). Fixture lists and club identifiers for past seasons come from a public archive of the same data (Anand, 2026). For every player and gameweek we record:

- minutes, starts, goals, assists, clean sheets, goals conceded and saves;
- bonus points and the bonus points system score;
- expected goals (xG), expected assists (xA) and expected goals conceded;
- total FPL points.

Expected goals estimate the probability that a shot is scored from its location and context (Rathke, 2017); FPL publishes Opta's values. In a double gameweek, when a club plays twice, the two matches are summed. Players and clubs keep the same identifier across seasons. Players with no minutes in a season are excluded for that season (Table 1). Because fixtures are known, a week with no fixture (a blank gameweek) is distinguished from a week in which a player did not play.

Table 1: Player-gameweek records by season.

Season	Player-gameweeks	Players	Total FPL points
2022/23	19,501	554	31,724
2023/24	20,526	570	31,273
2024/25	20,215	562	31,291
2025/26	19,643	537	34,382
Total	79,885		

Weekly prices were not archived, so each player carries his start-of-season price. The 2025/26 scoring rules added points for defensive actions (Fantasy Premier League, 2026b), which partly explains that season's higher total.

2.2 Squads of top-ranked managers

For 2023/24 to 2025/26 we hold the weekly fifteen-player squads, starting elevens, bench order, captain and vice-captain of the 100 managers who finished each season in the overall top 100: $3 \times 38 \times 100 = 11{,}400$ squad-weeks. Their most-captained player is a baseline, and their squads are a realistic test bed for lineup decisions. The sample is selected on the outcome: these managers are included *because* their season went well. Selecting on survival inflates apparent performance, a problem documented for investment funds (Brown et al., 1992), and it matters here (Section 6).

2.3 Season event data

For the usage analysis (Section 5) we use Opta season totals for every Premier League player from 2021/22 to 2025/26, published through the Premier League's statistics services (Premier League, 2026). Each player is assigned to the club he played for that season, taken from the season's squad lists, or, for players who left mid-season, from the club where he played most of his minutes.

3 Method

3.1 Notation and the information constraint

Index players by i, seasons by s and gameweeks by g, and write $t = (s, g)$ for a point in time ordered lexicographically. Let y_{it} be player i's points in week t, m_{it} his minutes and $n_{it} \in \{0, 1, 2\}$ the number of fixtures his club plays. The forecast for week t may use only weeks $t' < t$, together with quantities fixed before the deadline (position, start-of-season price, the fixture list, and the previous season's totals). We test this by perturbing the outcomes of one week and confirming that features for that week and all earlier weeks are unchanged.

3.2 Features

Player state is updated after each week in which the player's club had a fixture; the features for week t take the most recent state from a strictly earlier week. Table 2 lists each design choice with its basis.

Form and role. Minutes per fixture in the last week and averaged over the last three and six club-fixture weeks; the share of fixtures started over the last six; points per fixture over the last three and six. Short rolling windows of recent matches are the common input in FPL forecasting (Frees et al., 2024; Lombu et al., 2024; Ramezani and Dinh, 2025).

Per-90 rates. Over the last six club-fixture weeks $\mathcal{W}_6$, for a statistic z (xG, xA, bonus, bonus points system score, saves),

$$r_{it}^{(6)}(z) = 90 \cdot \frac{\sum_{t' \in \mathcal{W}_6} z_{it'}}{\max\left(\sum_{t' \in \mathcal{W}_6} m_{it'},\, 90\right)},$$

where the floor of 90 minutes stops a short cameo from producing an extreme rate.

Season to date. The same rates for xG, xA and points accumulated over the current season; the share of available minutes played, $\sum m / (90 \sum n)$; and the number of fixtures so far. These are missing in a season's first week.

Prior season. The player's xG, xA and points per 90, and his share of the season's minutes, in the previous season. Early in a season the current sample is a few matches and observed rates are dominated by noise. Combining them with a longer-run prior is the principle behind empirical Bayes estimation, which in sport outperforms the raw early-season rate (Brown, 2008; Efron and Morris, 1975). Here the trees learn how much weight the prior deserves at each point of the season; Section 4.4 tests whether it helps.

Club and opponent strength. For club c in week t, let $G^+_{ct} = \sum_{i \in c} \mathrm{xG}_{it}$ be its xG and $G^-_{ct} = \sum_o G^+_{ot}/n_{ot}$ its xG conceded, summing over its opponents o that week. Over the previous eight gameweeks, the club's attacking and defensive strength are

$$A_{ct} = \frac{\sum_{t'} G^+_{ct'}}{\sum_{t'} n_{ct'}}, \qquad D_{ct} = \frac{\sum_{t'} G^-_{ct'}}{\sum_{t'} n_{ct'}}.$$

Separate attack and defence parameters per club are the basis of the standard goal models for football (Dixon and Coles, 1997; Maher, 1982). We estimate them from xG rather than goals because xG is less noisy over short windows (Rathke, 2017). The player receives his club's A and D, the average of his opponents' A and D in week t, the number of fixtures, and the share played at home.

Static. Position, start-of-season price, gameweek number, an indicator for the defensive-action scoring rules, appearances to date, and the number of weeks since his last club-fixture week.

Table 2: Design choices and their basis.

Choice	Basis
Previous-season rates as a prior	Shrinkage of noisy early-season rates (Brown, 2008; Efron and Morris, 1975); no average gain in ablation (Table 5)
Rolling windows of recent matches	Standard in FPL forecasting (Frees et al., 2024; Lombu et al., 2024; Ramezani and Dinh, 2025)
Club attack/defence strength	Goal models with team attack and defence parameters (Dixon and Coles, 1997; Maher, 1982); xG as a lower-noise input (Rathke, 2017)
Gradient-boosted trees	Friedman (2001); histogram variant (Ke et al., 2017); used for FPL by Frees et al. (2024)
Model settings	Fixed before evaluation; sensitivity reported in Table 5
Walk-forward evaluation	Rolling-origin out-of-sample testing (Bergmeir and Benítez, 2012; Tashman, 2000)
Paired comparison by week	Comparing forecasts on common outcomes (Diebold and Mariano, 1995); weeks resampled by bootstrap (Efron and Tibshirani, 1993)
Acceptance rule set in advance	Pre-registration (Nosek et al., 2018)
Usage rate	Basketball usage percentage (Basketball-Reference, 2026; Oliver, 2004); football adaptation (Sincere FC, 2025a,b)

3.3 Model

The target is y_{it}, the player's total points in week t. We fit a gradient-boosted ensemble of regression trees (Friedman, 2001) with histogram binning of the inputs (Ke et al., 2017), as implemented in scikit-learn (Pedregosa et al., 2011), minimising squared error. The settings are:

- learning rate 0.05 and at most 250 boosting iterations, stopped early when the loss on a random 10% of the training window stops improving for ten iterations (the library's default whenever there are more than 10,000 training rows);

- at most 31 leaves per tree;
- at least 80 observations per leaf;
- an L_2 penalty of 1.0.

They start from the library defaults (learning rate 0.1, 100 iterations, 20 observations per leaf, no penalty) and move each setting in the conservative direction suited to a noisy target. They were fixed before evaluation and not tuned on the test seasons; Section 4.4 reports the effect of changing them.

Missing inputs, such as prior-season rates for a player new to the league, are handled natively by the trees. Features that are constant or entirely missing within a training window are dropped for that fit. A player with no previous club-fixture weeks receives no forecast. Training uses every player-week with a fixture and at least one prior observation. Squared error makes the forecast an estimate of the conditional mean, the right target for a captain choice that maximises expected points.

3.4 Decision rules

Captain. In each test week, the model captains the player with the highest forecast among those whose club has a fixture.

Lineup. Given a fifteen-player squad, the model chooses the legal formation that maximises the sum of forecasts. A formation has one goalkeeper, d defenders, m midfielders and f forwards, with $d \in \{3, 4, 5\}$, $m \in \{2, \ldots, 5\}$, $f \in \{1, 2, 3\}$ and $d + m + f = 10$. The two highest forecasts in the eleven are captain and vice-captain. The bench lists the reserve goalkeeper first and the outfield players by forecast.

Scoring. Every strategy is scored by the game's ordinary rules (Fantasy Premier League, 2026b). A starter with no minutes is replaced by the first bench player, in bench order, who played and whose entry keeps the formation legal (goalkeeper for goalkeeper; at least three defenders, two midfielders and one forward). The captain's points are doubled; if he did not play, the vice-captain's are. Chips are ignored: a triple captain counts as an ordinary captain and a bench boost as an ordinary week.

3.5 Baselines

All baselines use only information available at the deadline.

Top-100 consensus. The player most often captained that week by the 100 managers who finished the season in the overall top 100. This is the collective judgement of strong managers (Surowiecki, 2004), but the group is defined after the fact.

Premium attacker. The most expensive midfielder or forward (by start-of-season price) who played at least 60 minutes per fixture in his previous club-fixture week. Ties are broken by points per fixture over the last six.

Recent form. The highest points per fixture over the last six, among players averaging at least 60 minutes per fixture over the last three.

For lineups, the baselines are the manager's actual eleven, bench order and captain, and the model's own selection procedure ranking players by recent form.

3.6 Evaluation

For each test week w the model is retrained on all weeks before w, including earlier weeks of the same season, and used to forecast week w. This rolling-origin design mirrors live use (Tashman, 2000) and avoids the leakage that random cross-validation introduces in time-ordered data (Bergmeir and Benítez, 2012). The test period is the 114 gameweeks of 2023/24, 2024/25 and 2025/26.

Comparisons are paired by week, so that two rules are judged on the same outcomes (Diebold and Mariano, 1995). For rule A and baseline B, let $d_w = \text{pts}_w(A) - \text{pts}_w(B)$; for lineups, pts_w is first averaged over the 100 squads in week w. Weeks are the unit of resampling, since players and squads within a week share the same matches. We report $\bar{d}$ with a 95% percentile interval from 10,000 bootstrap resamples of the 114 weekly differences (Efron and Tibshirani, 1993).

Acceptance rule. Stated before the first evaluation: the model's captain choice is accepted only if the lower end of the interval is above zero against *every* captain baseline. The same rule applies to lineups. The absence of a detectable difference is not treated as evidence of parity.

Forecast accuracy. We report the mean absolute error (MAE) and the Pearson correlation of forecasts with outcomes over all test player-weeks with a fixture. We compare against a form forecast: points per fixture over the last six or, where unavailable, the prior season's points per 90 scaled by the player's share of minutes.

4 Results

4.1 Forecast accuracy

Over 58,682 test player-weeks, the model's MAE is 1.42 points against 1.50 for recent form, and its correlation with outcomes is 0.51 against 0.41 (Table 3). Most player-weeks score two points or fewer, so the correlation is the more informative comparison of ranking ability.

Table 3: Forecast accuracy over all test player-weeks with a fixture (2023/24–2025/26).

	Mean absolute error (points)	Correlation with outcome
Model	1.42	0.51
Recent form	1.50	0.41

4.2 Captaincy

Table 4 gives each rule's captain points per week (single, not doubled), and Figure 1 the paired differences.

- The model averages 6.75 points per week.
- It outscores the premium-attacker rule by +1.54 (95% interval +0.45 to +2.61) and the recent-form rule by +1.44 (+0.27 to +2.62).
- It trails the top-100 consensus captain by −1.46 (−2.40 to −0.55), so under the stated rule it is *not accepted*.
- The best possible captain each week scored 17.79 points on average, a mean regret of 11.04 points.

In 2023/24 the premium-attacker rule outscored the model (6.05 against 5.21); only in the two later seasons does the model clearly lead the mechanical rules. The top-100 consensus leads in every season.

Table 4: Captain points per gameweek (single) by season.

Season	Model	Top-100 consensus	Premium attacker	Recent form
2023/24	5.21	7.29	6.05	4.87
2024/25	8.82	10.13	5.26	6.71
2025/26	6.21	7.18	4.29	4.34
All (114 weeks)	6.75	8.20	5.20	5.31

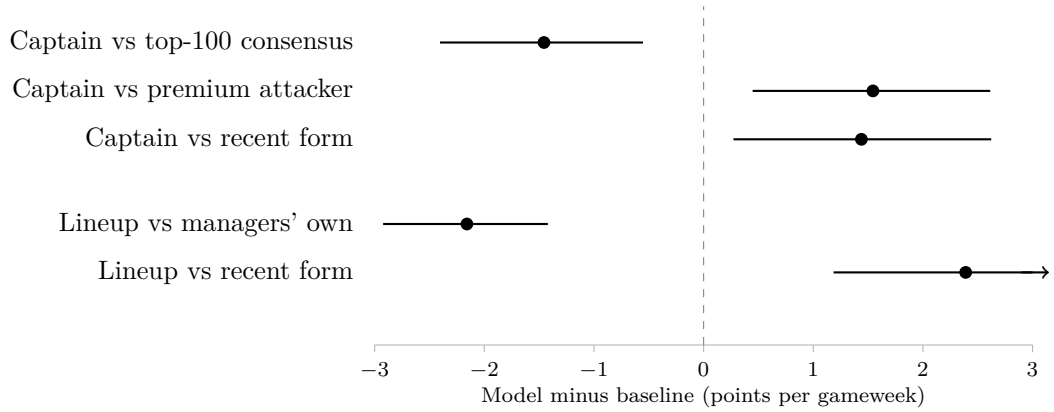


Figure 1: Paired weekly differences with 95% bootstrap intervals over 114 gameweeks. Points to the right of zero favour the model. The upper end of the last interval (+3.61) lies beyond the axis.

4.3 Lineups

On the 11,400 top-100 squad-weeks, the model's elevens and captains score 66.13 points per squad-week, against 68.29 for the managers' own choices and 63.74 for the form rule. The model beats the form rule by +2.39 (+1.19 to +3.61) and trails the managers by −2.16 (−2.92 to −1.42); it is not accepted for lineups either.

4.4 Which inputs matter, and how fragile are the settings?

Table 5 repeats the full walk-forward evaluation with one change at a time: a block of features removed, or one model setting moved. All runs, including the headline results, use the same software environment (scikit-learn 1.9.1, pandas 3.0.6, NumPy 2.4.6).

Three findings stand out.

- *Accuracy depends most on minutes and starts.* Removing them raises MAE from 1.415 to 1.442 and lowers correlation from 0.508 to 0.488, the largest loss of any block. Club and opponent strength (1.427, 0.501) and season-to-date rates (1.424, 0.502) come next.
- *Last season's rates add nothing on average.* Removing them leaves MAE and correlation unchanged (1.415, 0.508). The motivation for including them, stabilising early-season estimates, is not borne out over whole seasons. Carrying the rolling windows across season boundaries and the start-of-season price may already supply that information. Whether the prior helps in the first weeks alone is not tested here.

Table 5: Ablations and sensitivity. MAE and correlation over all test player-weeks; captain points per week; and the captain's mean weekly difference from the top-100 consensus and premium baselines. Each row is a full walk-forward run over 2023/24–2025/26.

	MAE	Corr.	Captain	vs top-100	vs premium
Reference model	1.415	0.508	6.75	−1.46	+1.54
Without prior season	1.415	0.508	6.87	−1.33	+1.67
Without club and opponent strength	1.427	0.501	6.75	−1.46	+1.54
Without per-90 rates (last six)	1.417	0.506	6.25	−1.95	+1.05
Without season to date	1.424	0.502	7.02	−1.18	+1.82
Without minutes and starts	1.442	0.488	6.79	−1.41	+1.59
Learning rate 0.1, 125 iterations	1.418	0.504	6.39	−1.81	+1.19
15 leaves per tree	1.415	0.508	6.90	−1.30	+1.70
63 leaves per tree	1.418	0.505	7.10	−1.10	+1.90
20 observations per leaf	1.419	0.505	6.57	−1.63	+1.37
Library defaults	1.422	0.502	6.34	−1.86	+1.14

- *The conclusions do not depend on the settings.* Across all eleven variants, MAE stays within 0.027 and correlation within 0.020 of the reference. Captain points range from 6.25 to 7.10 per week, but every variant still beats the premium-attacker rule (lower interval bounds +0.02 or above) and still trails the top-100 consensus (upper bounds −0.22 or below). The acceptance verdict is the same in every row.

The spread in captain points is itself informative. A preliminary run of the same variants under an earlier library version moved several of them in the opposite direction (for example, 63 leaves per tree gave 6.50 there and 7.10 here). Differences of half a point per week in captain points are within the noise of three seasons (Section 6). They should not be used to choose settings, and we have not done so.

5 Extension: ball usage

5.1 Motivation and definition

In basketball, usage percentage estimates the share of a team's possessions a player ends with a shot, free throws or a turnover while he is on the floor (Basketball-Reference, 2026). It builds on the possession-based framework of Oliver (2004). It separates how much of the ball a player consumes from what he produces with it. A recent football adaptation (Sincere FC, 2025a,b) counts *possession-ending actions*: shots, and every way of losing the ball (an incomplete pass or cross, a failed take-on, being tackled or dispossessed, a miscontrol). It argues that the measure describes a player's role and hierarchy within a team. A player used to being his team's main option may struggle to produce when he joins a club that already has one. If so, last season's usage should carry information about this season's output, especially for players who change club, where the prior-season rates in our model are least reliable.

We define a player's possession-ending actions as Opta's possessions lost plus his shots. Possessions lost covers incomplete passes and crosses, failed take-ons, being tackled and miscontrols, and a little more: for Premier League players with at least 1,500 minutes in 2023/24–2025/26 it is a median 8% larger than the sum of those itemised components (middle 90% of players: 1% to 32% larger). Following the basketball formula with eleven players in place of five,

$$\mathrm{USG}_i = 100 \cdot \frac{\mathrm{PEA}_i \cdot (\text{team minutes}/11)}{\text{minutes}_i \cdot \mathrm{PEA}_{\text{team}}},$$

where team totals are taken over every player of the club he played for that season. For creation we use Opta's chances created: passes that lead directly to a shot. The adaptation above uses a broader measure of shot-creating actions, which our source does not publish.

5.2 Features and test

We add eight features computed from the *previous* season's totals, so all are known before the season being forecast:

- usage rate;
- possession-ending actions per 90;
- chances created per 90;
- chances created per possession-ending action;
- touches in the opposition box per 90;
- the club's possession share;
- an indicator that the player has changed club;
- the highest previous-season usage among his current teammates.

The last feature asks whether the club he now plays for already has a dominant ball user. We then rerun the walk-forward evaluation with and without these features on identical weeks, and bootstrap the paired weekly differences.

Table 6: Walk-forward evaluation with and without the usage features (2023/24–2025/26). Differences are with-usage minus without, paired by gameweek, with 95% bootstrap intervals. [a]Reduction in MAE (positive means the usage features help).

	Without usage	With usage	Difference [95% interval]
Mean absolute error	1.415	1.414	0.000 [−0.003, +0.003][a]
Correlation	0.508	0.507	
Captain points per week	6.75	6.47	−0.27 [−1.26, +0.69]
Lineup points per week	66.13	66.14	+0.00 [−0.71, +0.74]
MAE, players who changed club			+0.001 [−0.005, +0.006][a]

5.3 Result

The usage features were present for 57% of test player-weeks; the rest belong to players with no Premier League season the year before. Players who had changed club since the previous season account for 10% of test player-weeks. Table 6 shows no detectable effect:

- forecast error is unchanged (MAE reduction 0.000, 95% interval −0.003 to +0.003), including for players who changed club (+0.001, −0.005 to +0.006);
- captain points change by −0.27 per week (−1.26 to +0.69);
- lineup points are unchanged (+0.00, −0.71 to +0.74).

Over these three seasons, last season's ball usage adds nothing that last season's xG, xA, points and minutes do not already carry, so we do not adopt it. Two caveats apply. First, the measure is a season total; an in-season, match-level version would capture role changes as they happen. Second, usage may matter for questions this model does not ask, such as how a new signing's arrival redistributes the ball among his teammates, which needs club-level rather than player-level modelling.

6 Discussion

What the model has learned. The forecast ranks players better than recent points do: its correlation with outcomes is about a quarter higher. As a decision rule it beats both mechanical captain heuristics, and it picks better elevens than form does from the same squads. These gains come mainly from minutes security, club and opponent strength and current-season rates (Section 4.4), and from not over-reacting to a short run of high or low scores. Neither last season's rates nor last season's ball usage improved the forecast on average.

Why the top-100 consensus is hard to beat. The consensus baseline has two advantages. The first is genuine. At the deadline, managers know team news, late injuries, the likely penalty taker and the betting market's view of each fixture, none of which the model sees. Betting markets in particular are known to carry information beyond statistical models (Dixon and Coles, 1997), and adding news and market data has improved FPL forecasts in earlier work (Bonello et al., 2019). The second is an artefact of selection. These managers are in the sample because their season went well, so their captaincy choices are, on average, those that worked out (Brown et al., 1992). A fair comparison would use the consensus of a sample fixed before the season.

How much evidence a season provides. Inverting the reported intervals gives a standard deviation of the weekly paired captain difference of roughly 5 to 6 points: for the top-100 comparison, $\hat{\sigma} \approx 0.93\sqrt{114}/1.96 \approx 5.0$. With $\sigma = 5$, a 95% interval of half-width h requires about $(1.96\,\sigma/h)^2$ weeks. That is 96 weeks for $h = 1$ point and about 380 weeks for $h = 0.5$; a single season gives $h \approx 1.6$. Claims that one captaincy rule beats another by half a point per week cannot be supported by one season, or even three.

Limitations.

- *Missing information.* The model has no team news, injury history, set-piece duties or bookmaker prices.
- *Prices.* Only start-of-season prices are available historically.
- *Selection on the outcome.* The top-100 sample is selected on how the season turned out.
- *Serial dependence.* Weekly differences may be serially dependent within a season; a block or stationary bootstrap (Künsch, 1989; Politis and Romano, 1994) would widen the intervals somewhat.
- *Reuse of the test seasons.* The usage extension and any further development evaluated on these same three seasons risk overstating performance; improvements should be confirmed prospectively.

Prospective evaluation. From the seventh gameweek of 2026/27, the following captains are logged before each deadline:

- the model's;
- the one implied by the official game's expected-points figure;
- the premium-attacker rule's.

The most-captained player across all managers is recorded once the gameweek begins. Logs are frozen at the deadline and scored when the gameweek finishes. The official expected points, ownership and availability flags are stored at each deadline. This closes part of the information gap for future model versions and gives an out-of-sample record that cannot be revised with hindsight.

7 Conclusion

A gradient-boosted forecast trained on four seasons of public data makes better captain and lineup choices than simple rules, and ranks players better than recent form. It does not yet match the collective judgement of the season's best managers, whose advantage reflects both information the model lacks and the way they were selected. Distinguishing good rules from slightly better ones requires many more weeks than a season provides. Evaluation should therefore be prospective, pre-specified and patient.

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A Features

Table 7: Model inputs. "Club-fixture week" means a gameweek in which the player's club played; all windows end strictly before the forecast week. The usage block is used only in the extension of Section 5.

Group	Features
Static	position; start-of-season price; gameweek number; defensive-action scoring era (0/1)
Fixture	number of fixtures in the week; share at home; own club's rolling xG for and against; opponents' rolling xG for and against (eight-gameweek windows)
Form	minutes per fixture (last 1, mean of last 3, mean of last 6); share of fixtures started (last 6); points per fixture (last 3, last 6)
Per-90 rates (last 6)	xG, xA, bonus points, bonus points system score, saves
Season to date	xG, xA and points per 90; share of available minutes played; fixtures so far
Prior season	xG, xA and points per 90; share of the season's minutes
Exposure	appearances to date; weeks since the last club-fixture week
Usage (extension)	previous season's usage rate, possession-ending actions per 90, chances created per 90 and per action, box touches per 90, club possession; changed club (0/1); highest previous-season usage among current teammates